



Islamic University of Technology
Department of Computer Science and Engineering

Lab 2: Sorting

CSE 4404: Algorithms Lab(SWE)
Summer 2024-25

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Task A. Radix Sort the Array

Time Limit: 1 second | Memory Limit: 256 MB

You are given an array of n non-negative integers. Your task is to sort the array in non-decreasing order and print the result.

You must implement **Radix Sort** using a counting array to achieve this. Using any comparison-based sorting algorithm (such as `qsort`, `std::sort`, bubble sort, merge sort, etc.) will not be accepted.

Input Format

The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases.

For each test case:

- The first line contains a single integer n ($1 \leq n \leq 10^5$) — the number of elements.
- The second line contains n space-separated integers $a_1, a_2, \dots, a_n$ ($0 \leq a_i \leq 10^9$) — the elements of the array.

The sum of n over all test cases does not exceed 10^5 .

Output Format

For each test case, print a single line containing n space-separated integers — the array sorted in non-decreasing order.

Examples

Sample Input	Sample Output
4	1 2 3 4 5
5	7
5 3 1 4 2	0 0 0 0 0 0
1	0 1 999999999 1000000000
7	
6	
0 0 0 0 0 0	
4	
1000000000 999999999 1 0	

Task B. Weighted Difference Minimization

Time Limit: 1 second | Memory Limit: 256 MB

You are given three arrays a , b , and w , each containing n integers.

You have to rearrange the elements of all three arrays in any order so that the following value is minimized:

$$\sum_{i=1}^n w_i \cdot |a_i - b_i|$$

That is, after permuting the arrays, choose an arrangement such that the weighted sum of absolute differences between corresponding elements of a and b is as small as possible.

Input Format

The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases.

For each test case:

- The first line contains a single integer n ($1 \leq n \leq 2 \times 10^5$) — the size of the arrays.
- The second line contains n integers $a_1, a_2, \dots, a_n$ ($|a_i| \leq 10^9$).
- The third line contains n integers $b_1, b_2, \dots, b_n$ ($|b_i| \leq 10^9$).
- The fourth line contains n integers $w_1, w_2, \dots, w_n$ ($1 \leq w_i \leq 10^9$).

It is guaranteed that the sum of n over all test cases does not exceed 2×10^5 .

Output Format

For each test case, print a single integer — the minimum possible value of $\sum_{i=1}^n w_i \cdot |a_i - b_i|$ after optimal rearrangement.

Examples

Sample Input	Sample Output
3	25
3	0
1 7 10	115
2 8 20	
5 1 10	
4	
1 2 3 4	
4 3 2 1	
1 1 1 1	
5	
10 10 10 10 10	
1 2 3 4 5	
5 4 3 2 1	

Task C. Minimum Bottles to Reach Target

Time Limit: 1 second | Memory Limit: 256 MB

You are given n empty bottles, each with a certain capacity. You also have access to an unlimited water source.

Your goal is to select the minimum number of bottles such that the total capacity of the selected bottles is at least x liters.

Each selected bottle must be filled completely.

Input Format

The first line contains a single integer t ($1 \leq t \leq 10^5$) — the number of test cases.

For each test case:

- The first line contains two integers n and x ($1 \leq n \leq 10^4$, $1 \leq x \leq 10^9$) — the number of bottles and the required amount of water.
- The second line contains n space-separated integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^9$), where a_i is the capacity of the i -th bottle.

It is guaranteed that the sum of n over all test cases does not exceed 10^5 .

Output Format

For each test case, print a single integer:

The minimum number of bottles required such that their total capacity is at least x .

If it is impossible, print -1 .

Examples

Sample Input	Sample Output
3	3
5 10	2
2 3 4 5 1	3
4 7	
5 3 4 2	
3 20	
6 7 8	

Task D. Count Values in Range

Time Limit: 2 seconds | Memory Limit: 512 MB

You are given an array of n integers. You also have to answer q queries.

For each query, you are given two integers l and r , and you must determine how many indices i ($1 \leq i \leq n$) satisfy the condition $l \leq a_i \leq r$.

Input Format

The first line contains two space-separated integers n and q ($1 \leq n, q \leq 10^5$) — the size of the array and the number of queries.

The second line contains n space-separated integers $a_1, a_2, \dots, a_n$ ($-10^9 \leq a_i \leq 10^9$) — the elements of the array.

Each of the next q lines contains two space-separated integers l and r ($-10^9 \leq l \leq r \leq 10^9$) — describing a query.

Output Format

For each query, print a single integer — the number of indices i such that $l \leq a_i \leq r$.

Examples

Sample Input	Sample Output
5 3	4
1 5 2 7 3	3
1 5	0
2 6	
8 10	

Notes

In the first query, the indices satisfying $1 \leq a_i \leq 5$ are $i = 1, 2, 3$, and 5 — a total of 4 indices.

In the second query, the indices satisfying $2 \leq a_i \leq 6$ are $i = 2, 3$, and 5 — a total of 3 indices.

In the third query, no index satisfies $8 \leq a_i \leq 10$.

Marks Distribution

Task	Marks
Task A	25%
Task B	25%
Task C	25%
Task D	25%

Practice Problems

Problem	Links
Game	Codeforces: https://codeforces.com/contest/984/problem/A Vjudge: https://vjudge.net/problem/CodeForces-984A
Gravity Flip	Codeforces: https://codeforces.com/contest/405/problem/A Vjudge: https://vjudge.net/problem/CodeForces-405A
Valid Anagram	LeetCode: https://leetcode.com/problems/valid-anagram/
Jagged Swaps	Codeforces: https://codeforces.com/contest/1896/problem/A Vjudge: https://vjudge.net/problem/CodeForces-1896A
Unforgivable Curse (Easy)	Codeforces: https://codeforces.com/contest/1800/problem/E1 Vjudge: https://vjudge.net/problem/CodeForces-1800E1
Unforgivable Curse (Hard)	Codeforces: https://codeforces.com/contest/1800/problem/E2 Vjudge: https://vjudge.net/problem/CodeForces-1800E2